If the pre-trade estimates are accurate, then the z-score statistic should be a random variable with mean zero and variance equal to one. That is, $Z \sim (0, 1)$. There are various statistical tests that can be used to test this joint hypothesis.

There are several points worth mentioning with regards to trading cost comparison. First, the test needs to be carried out for various order sizes (e.g., large, small, and mid-size orders). It is possible for a model to overestimate costs for large orders and underestimate costs for small orders (or vice versa) and still result in $Z \sim (0, 1)$ on average. Second, the test needs to be carried out for various strategies. Investors need to have a degree of confidence regarding the accuracy of cost estimates for all of the broker strategies. Third, it is essential that the pre-trade cost estimate be based on the number of shares traded and not the full order. Otherwise, the pre-trade cost estimate will likely overstate the cost of the trade and the broker being measured will consistency outperform the benchmark giving the appearance of superior performance and broker ability. In times where the order was not completely executed, the pre-trade cost estimates need to be adjusted to reflect the actual number of shares traded. Finally, analysts need to evaluate a large enough sample size in order to achieve statistical confidence surrounding the results, as well as conduct cross-sectional analysis in order to uncover any potential bias based on size, volatility, market capitalization, and market movement (e.g., up days and down days).

It is also important to note that many investors are using their own pre-trade estimates when computing the z-score measure. There is a widespread resistance to using a broker's derived pre-trade estimate to evaluate their own performance. As one manager stated, everyone looks great when we compare their performance to their cost estimate. But things start to fall into place when we use our own pre-trade estimate. Pre-trade cost comparison needs to be performed using a standard pre-trade model to avoid any bias that may occur with using the provider's own performance evaluation model.

Market Cost Adjusted Z-Score

It is possible to compute a z-score for the market adjusted cost as a means of normalizing performance and comparing across various sizes, strategies, and time periods similar to how it is used with the trading cost metric. But in this case, the denominator of the z-score is not the timing risk of the trade since timing risk accounts in part for the uncertainty in total price movement (adjusted for the trade schedule). The divisor in this case has to be the tracking error of the stock to the underlying index (adjusted for